C-Band 6×6 Dual-Polarised Antenna Array

Input-match (VSWR) investigation: root cause, findings and corrective actions

Document	TM-RF-15-001, Revision A
Date	30 October 2026
Report type	Technical investigation report
Frequency band	4.0–7.0 GHz (C-band)
Prepared by	Telmart RF & Microwave Engineering
Status	Interim: full-array verification in progress

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1 Executive summary

The 6×6 dual-polarised stacked-patch array radiates with the expected broadside gain, but its input match is poor: the VSWR does not stay below 3:1 across the 4–7 GHz band. Telmart analysed the design with full-wave electromagnetic simulation built directly from the supplied layout files, at element level and for the complete array.

~16 %
Element bandwidth (VSWR ≤ 3), 5.35–6.26 GHz

54 %
Bandwidth required for 4–7 GHz

8.7–9.8 dBi
Element gain referred to accepted power: the radiator works

VSWR ≈ 9
Best achievable with any feed-network match (worst case 4–7 GHz)

Root cause. The radiating element is inherently narrow-band. The edge-fed driven patch and the parasitic patch are nearly the same size and produce a single merged resonance near 5.8 GHz, instead of two separate resonances spread across the band. Outside about 5.35–6.26 GHz most of the power is reflected. That is why the gain looks correct while the VSWR is poor.

Conclusion. The problem cannot be fixed in the feed network alone. Meeting VSWR < 3 over 4–7 GHz requires a redesigned wideband radiating element. Short-term improvements (equal parasitic sizes, per-element transformers, assembly tolerances) centre and stabilise the present band but will not widen it to 4–7 GHz.

Status. The element-level analysis is complete. The full-array simulation is still running; the preliminary array results in section 5.4 agree with the element-level conclusions, and bench measurements of the built panel (section 5.6) confirm the input-match behaviour. The measurements also show a gain shortfall of about 3 dB mid-band and a strong gain drop above 6 GHz, which is now part of the investigation. A first wideband redesign iteration has been completed and has defined the requirements for the next design iteration (section 7).

2 Reported issue and scope

Item	Description
Symptom	Broadside gain as expected; input VSWR above 3:1 over most of 4–7 GHz on both polarisations
Product	6×6 dual-linear-polarised stacked-patch array, 276.86 × 276.86 mm, two coaxial inputs (V-pol and H-pol)
Objective	Identify the root cause of the poor match, assess whether it can be corrected, and define corrective actions
Inputs received	Layer files F.Cu (parasitic patches), In2.Cu (driven patches and feed networks), B.Cu (ground) and the stack-up drawing
Out of scope	Measurement on the physical board, connector and mechanical spacer modelling, production tolerance analysis

3 Design under review

The array uses 36 identical driven patches (17.07 mm) on a 0.8 mm RT5880 laminate. Each driven patch has a parasitic patch (16.2–17.8 mm, graded by row) on 0.4 mm FR4, held 4 mm above it by an air gap. Two independent 36-way corporate networks on the driven-patch layer feed the top/bottom edges (port 1, V-pol) and the left/right edges (port 2, H-pol) of every patch.

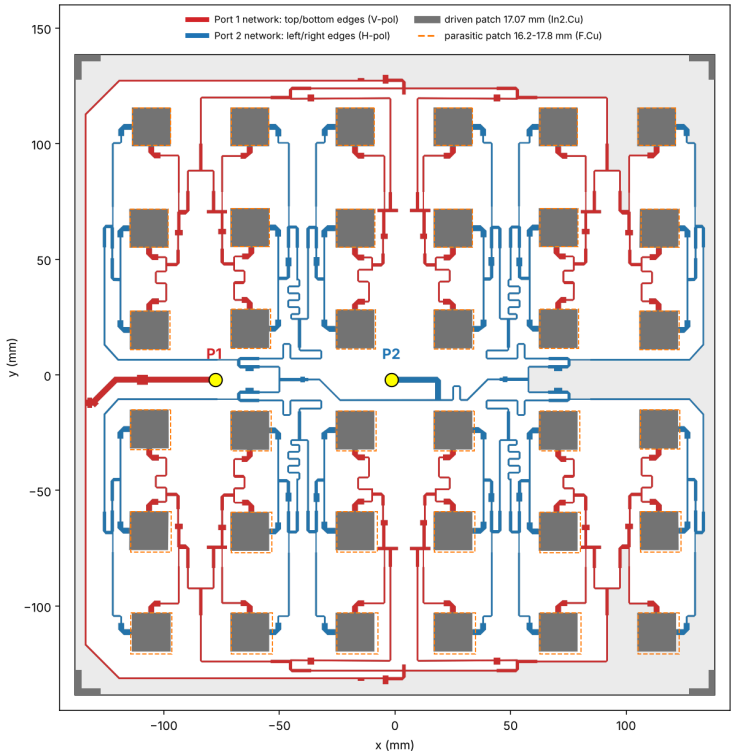


Figure 1. Driven-patch layer with the two feed networks highlighted (red: port 1 / V-pol, blue: port 2 / H-pol). Dashed outlines are the parasitic patches.

Parameter	Value
Element pitch	≈ 44 mm (0.59 λ at 4 GHz, 1.03 λ at 7 GHz)
Stack-up	Ground / RT5880 0.8 mm / driven patches + feeds / 4 mm air / FR4 0.4 mm / parasitic patches
Feed lines	0.73 mm (100 Ω), 1.38 mm (70.7 Ω quarter-wave transformers), 2.48 mm (50 Ω)
Element feed	Direct edge feed through a short 50 Ω stub, no matching section

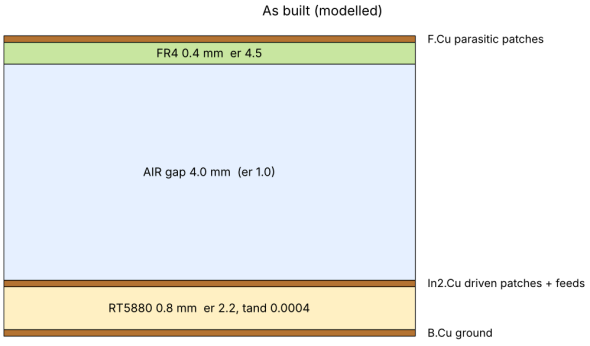
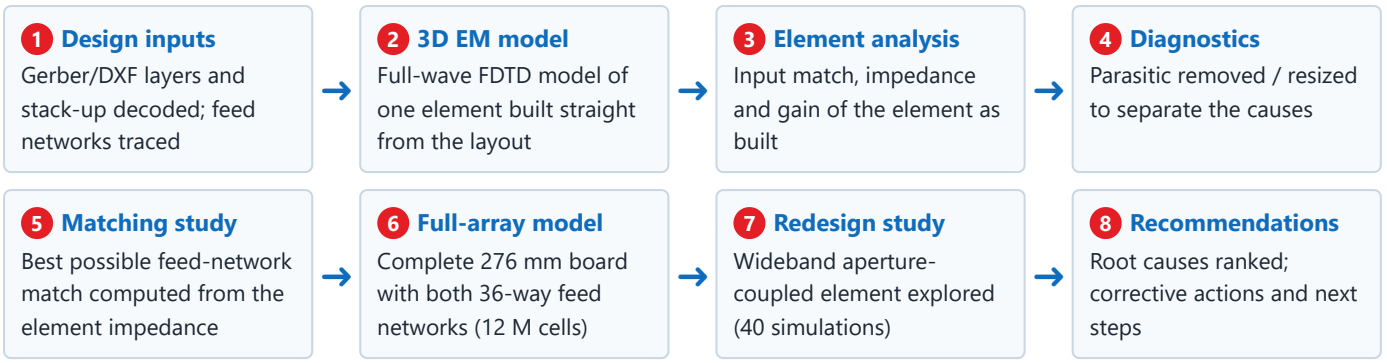


Figure 2. Stack-up as built and simulated.

4 Investigation process

The investigation followed a cause-isolation approach: first establish what one element does on its own, then rule out or confirm the feed network, and finally confirm on the complete array.



Method	Details
Solver	Full-wave FDTD electromagnetic simulation (openEMS), 2.5–8.5 GHz excitation, absorbing boundaries, near-to-far-field transform for gain
Geometry	Imported directly from the supplied layer files, no manual redrawing; copper as perfect conductor
Materials	RT5880 ε_r 2.2, tanδ 0.0004; FR4 ε_r 4.5, tanδ 0.02; air gap ε_r 1.0
Element model	One element with both feeds, 0.25 mm mesh, 2.9 M cells
Array model	Complete board with both networks and ports, 0.5 mm mesh, 12 M cells
Matching study	Optimised line / transformer / stub networks applied to the simulated element impedance

5 Findings

5.1 Element performance as built

A single element, simulated with its own feed stubs, is matched (VSWR ≤ 3) only from about 5.35 to 6.26 GHz, with a best return loss of 10.8 dB at 5.84 GHz. The element still radiates efficiently (8.7–9.8 dBi referred to accepted power), but the realised gain drops at the band edges because of the mismatch.

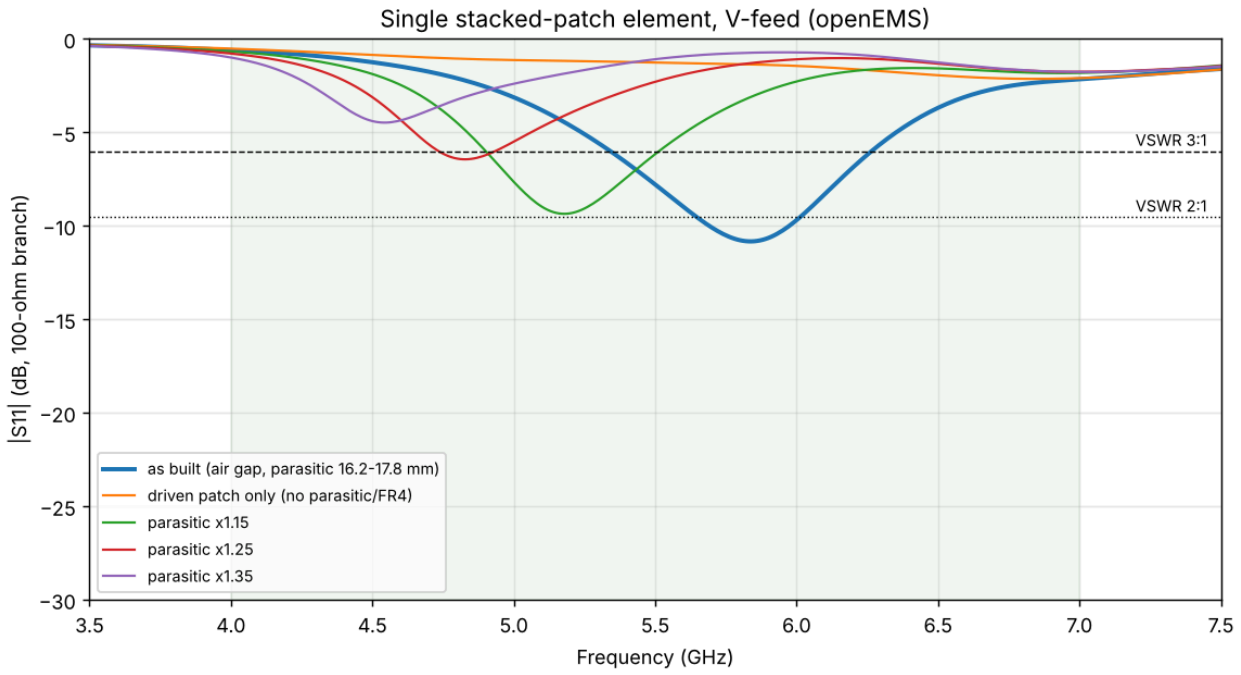


Figure 3. Element return loss: as built (bold) and diagnostic variants. Dashed line: VSWR 3:1; dotted: VSWR 2:1; shaded: 4–7 GHz target band.

5.2 Diagnostic variants

Variant	Result	What it shows
Parasitic patch removed	No match anywhere in 4–7 GHz	The edge feed is not matched to the patch; the match relies entirely on the parasitic
Parasitic enlarged ×1.15 / 1.25 / 1.35	Matched window moves down and becomes narrower (12 %, 4 %, none)	Resizing tunes *where* the element works, not *how wide*
Parasitic graded by row (as built)	Each row matched at a slightly different frequency	Part of the aperture is always mismatched

5.3 Can the feed network correct it?

The best possible passive match was computed from the simulated element impedance (any series line, quarter-wave transformer and shunt stub). The worst-case VSWR over 4–7 GHz improves only from 28.6 to about 9.4. The element's impedance varies too much across the band to be compensated by a passive network (Bode–Fano limit). **The correction must be made in the radiating element.**

5.4 Full array (preliminary)

The complete board, with both 36-way networks, is being simulated. Because the long feed networks ring for a long time, the run takes many hours. The preliminary results below are taken from the data recorded so far.

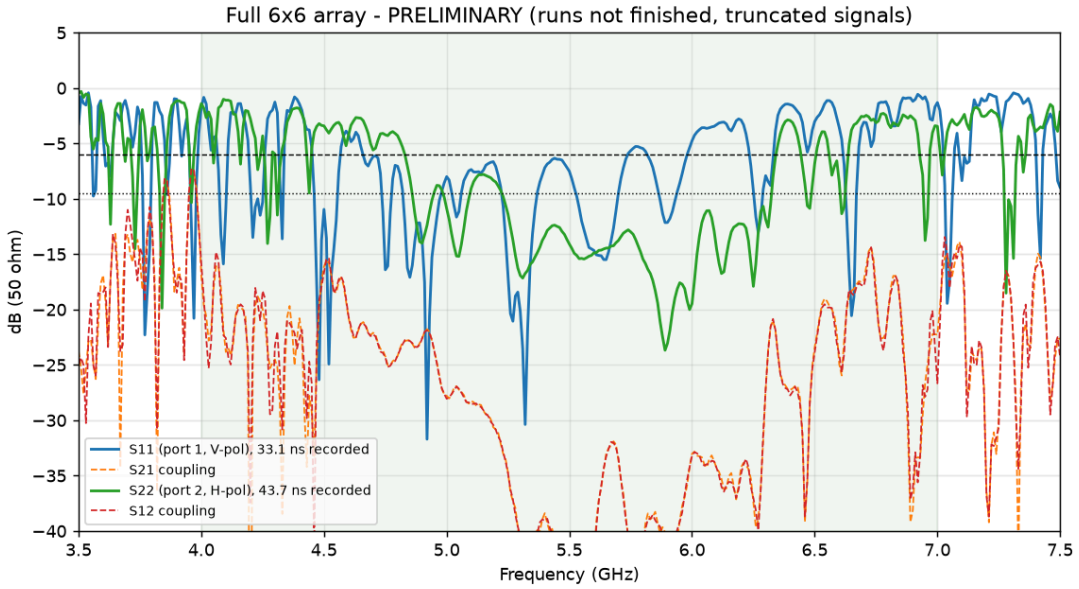


Figure 4. Preliminary full-array input reflection (S11 V-pol, S22 H-pol) and port-to-port coupling. Sharp dips and ripple are partly due to the run not yet being complete.

- Both inputs are best in the middle of the band (about 5.0–6.2 GHz) and mostly at VSWR 3:1 or worse below ~4.8 GHz and above ~6.3 GHz, matching the element result.
- The V/H isolation is better than 20 dB over most of the band.
- Final array figures will be issued in the next revision of this report.

5.5 Wideband redesign: first iteration

To define the corrective design, an aperture-coupled stacked patch, the established route to about 50 % bandwidth, was explored in 40 full-wave simulations, including automatic optimisation. The best candidate so far is matched from 4.95 to 5.75 GHz. It is a valid design but not yet wider than the present element. The iteration identified clearly what the next design must change:

- Patches must be printed on thin low-loss laminates over foam rather than suspended in air, so the two resonances can be separated across the band.
- A resonant (dog-bone or H-shaped) coupling slot is needed to add a third resonance at the top of the band; H-slots also provide the second polarisation.
- The element must be optimised in its array environment (periodic unit cell at the 44 mm pitch).

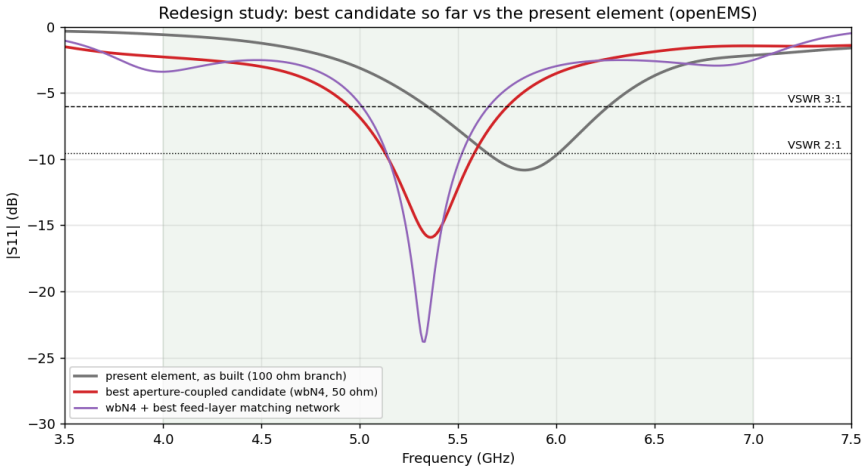


Figure 5. First-iteration wideband candidate compared with the present element.

5.6 Measured results of the built panel

The client's built panel was measured on the bench: input match of both ports with a calibrated network analyser, and broadside gain by substitution against a reference horn.

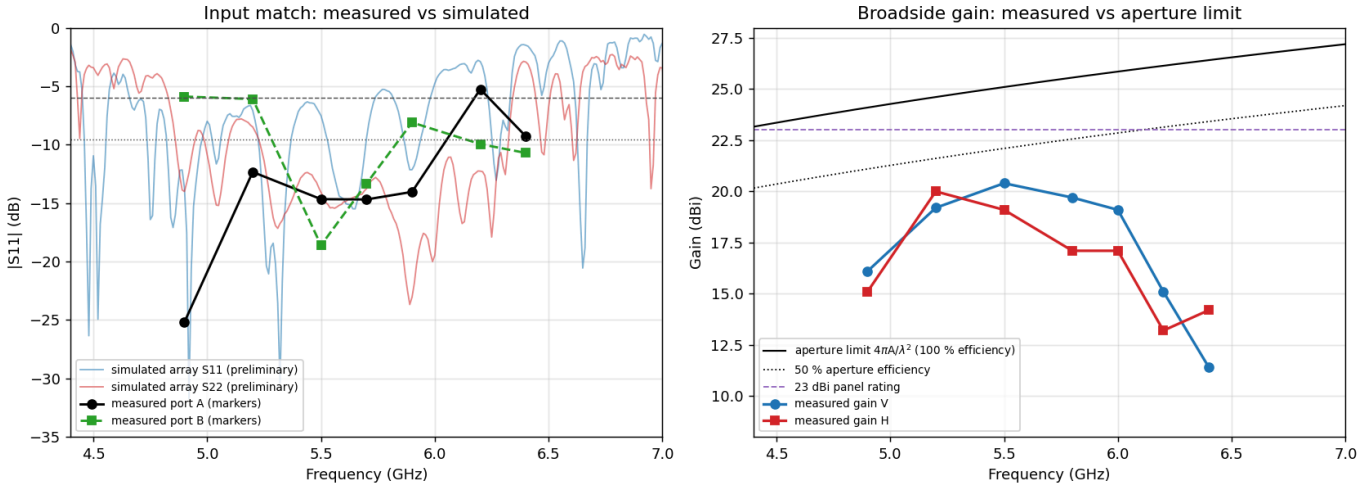


Figure 6. Measured panel against simulation. Left: measured input match (markers) over the simulated array curves. Right: measured gain against the aperture limit of the 276.86 mm panel and the 23 dBi rating.

f (GHz)	S11 port A (dB)	S11 port B (dB)	Gain V (dBi)	Gain H (dBi)	Aperture limit (dBi)	Aperture efficiency V / H
4.9	-25.2	-5.9	16.1	15.1	24.1	16 % / 13 %
5.2	-12.3	-6.1	19.2	20.0	24.6	29 % / 35 %
5.5	-14.7	-18.6	20.4	19.1	25.1	34 % / 25 %
5.8	-14.7*	-13.3*	19.7	17.1	25.6	26 % / 14 %
6.0	-14.0*	-8.1*	19.1	17.1	25.9	21 % / 13 %
6.2	-5.3	-9.9	15.1	13.2	26.1	8 % / 5 %
6.4	-9.3	-10.7	11.4	14.2	26.4	3 % / 6 %

* nearest S11 marker (5.7 / 5.9 GHz). Gain by substitution against a reference horn at ~5 m. Port A/B to V/H mapping and radome condition during the gain test to be confirmed.

- **Input match confirms the analysis.** Both ports show the same rippled response as the simulated array and are best in the middle of the band. Port A meets VSWR ≤ 3 at all markers except 6.2 GHz; port B is at about VSWR 3 at 4.9–5.2 GHz.
- **Gain is below the rating.** Peak gain is 20.4 dBi (V) and 20.0 dBi (H), about 3 dB under 23 dBi, corresponding to 25–35 % aperture efficiency mid-band (a well-designed panel reaches 50–70 %).
- **Gain collapses at the top of the band:** 11–15 dBi at 6.2–6.4 GHz (3–8 % efficiency). Mismatch explains at most about 1.5 dB of this. The remaining shortfall points to losses or phase errors (feed network, FR4 parasitic layer, radome) and is being investigated.

6 Root causes

#	Cause	Evidence	Impact
1	Element bandwidth (~16 %) far below the 54 % target	Element simulation, impedance locus, matching study	VSWR > 3 over most of 4–7 GHz, whatever the feed network
2	Driven patch edge-fed directly from a 50 Ω stub, without a matching section	No match at all without the parasitic	Match depends entirely on the parasitic loading
3	Driven and parasitic resonances merge (similar sizes)	Single resonance; resizing only shifts the window	No double-tuned wideband response
4	Parasitic sizes graded by row (16.2–17.8 mm)	Layout; parasitic-size sweep	Rows matched at different frequencies
5	Element pitch ≈ 1.03 λ at 7 GHz	Layout geometry	Grating lobes and gain reduction near 7 GHz (affects gain, not VSWR)
6	Gain shortfall and high-band gain drop (under investigation)	Bench measurement: 25–35 % aperture efficiency, 3–8 % at 6.2–6.4 GHz	About 3 dB below the 23 dBi rating mid-band; 6–9 dB further loss at 6.2–6.4 GHz

7 Recommendations and next steps

7.1 Corrective actions

#	Action	Expected result	Type
1	Redesign the radiating element as a wideband dual-polarised element (aperture-coupled stacked patch on laminate/foam, or L-probe / crossed-dipole alternative)	Only route to VSWR < 3 (target < 2) over 4–7 GHz	Required
2	Use one parasitic size on all rows unless a taper is intended	All rows matched at the same frequency	Quick fix
3	Add a quarter-wave transformer on each element feed (edge resistance to 100 Ω)	Present band centred and deeper; not wider	Quick fix
4	Control the 4 mm air-gap spacer tolerance (±0.1 mm)	Stable, repeatable match between boards	Production
5	Reduce the pitch to ≤ 30 mm if the full band must be free of grating lobes	Clean patterns up to 7 GHz (needs the smaller wideband element)	Design

7.2 Plan for the next phase

Phase	Activity	Deliverable
1	Complete the full-array simulation of the present design and compare its gain with the measurement	Final array S-parameters and gain; cause of the gain shortfall; report revision B
1b	Bench checks: gain and S11 with and without radome, without the parasitic layer, and pattern cuts at 5.5 and 6.4 GHz	Separates radome, parasitic-layer and feed-phase effects
2	Wideband element, iteration 2: unit-cell model, laminate/foam stack, resonant H-slots, dual polarisation, systematic optimisation	Element meeting VSWR < 3 (target < 2) over 4–7 GHz, with isolation and pattern data
3	Array integration: feed network redesign for the new element, full-array verification	Array layout files and simulated performance
4	Prototype and measurement: element test coupon, network with resistive loads, full array	Measured S-parameters and patterns; correlation with simulation

> **Recommended immediate bench check:** measure one element on a test coupon and the bare feed network loaded with 100 Ω resistors. This separates element and network on the bench and confirms the findings of this report on hardware.

8 Assumptions and limitations

- Bench measurements were made by the client team on a bench range (~5 m), not in an anechoic chamber; gain values carry an estimated ±1–2 dB uncertainty. The port A/B to V/H mapping and the radome condition during the gain test are to be confirmed.
- The air gap is modelled as 4.0 mm of air (ε_r 1.0), as built.
- Copper thickness, solder mask, connectors and the mechanical spacers are not modelled.
- The array mesh (0.5 mm) approximates the 0.73 mm lines, which can shift network resonances by a few percent.
- Full-array results in this revision are preliminary (simulation still running).